

Problem 1 (13 marks)

Two countries, **Home (H)** and **Foreign (F)**, produce two goods: **textiles (T)** (labour-intensive) and **steel (S)** (capital-intensive). Both countries have the same technologies and homothetic preferences.

Country	Capital K	Labour L	K/L ratio
Home	300	900	0.33
Foreign	800	800	1.00

Assume perfect competition, identical production functions, and full employment.

- (1) Identify which country is relatively labour-abundant and which is capital-abundant.
- (2) According to the Heckscher–Ohlin theorem, which good will each country export under free trade?
- (3) Using the Stolper–Samuelson theorem, explain how opening to trade affects the real wage and the real rental rate in each country.
- (4) Suppose the world relative price of textiles (P_T/P_S) rises by 10%. Explain how this change affects factor incomes (wages and rents) and the output composition in the Home country.
- (5) If trade continues over time, what mechanism could lead to **factor price equalization** between the two countries?